CBSE Class 10 Physics: Magnetic Effects of Electric Current

Complete Concept Notes, Hand Rules, Solved Examples & Exam Practice Questions

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1. Magnetism and Magnetic Field Lines

1.1 Oersted's Experiment & Magnetic Field

In 1820, Hans Christian Oersted discovered that an electric current flowing through a metallic conductor produces a magnetic field around it. A magnetic field is the region around a magnet or current-carrying conductor where its magnetic force can be detected.

- **SI Unit of Magnetic Field (B):** Tesla (T).
- **Vector Quantity:** Magnetic field has both magnitude and direction.

1.2 Magnetic Field Lines & Properties

Magnetic field lines are imaginary lines representing the direction and strength of the magnetic field.

- **Direction:** Outside the magnet, lines emerge from the **North pole** and enter the **South pole**. Inside the magnet, they move from **South to North** (forming closed continuous loops).
- **Field Strength:** Relative strength is indicated by the degree of closeness of field lines (closer lines = stronger field).
- **Non-Intersecting Property:** Two magnetic field lines **never cross each other**. If they did, a compass needle at the point of intersection would point in two different directions simultaneously, which is impossible.

2. Magnetic Field Due to Current-Carrying Conductors

2.1 Straight Conductor

The magnetic field lines around a straight current-carrying wire form **concentric circles** centered on the wire.

- Field magnitude (B) is directly proportional to current (I) and inversely proportional to distance (r) from the conductor.

Right-Hand Thumb Rule (Maxwell's Rule)

Imagine holding a current-carrying straight conductor in your right hand such that the thumb points in the direction of current. Then, your curled fingers wrap around the conductor in the direction of the magnetic field lines.

2.2 Circular Loop

At every point of a current-carrying circular loop, the magnetic field lines are concentric circles. Near the center of the loop, the arcs of these circles appear as **straight lines**.

- Magnetic field strength at the center increases directly with the number of turns (N) in the coil: $B \propto N$.

2.3 Solenoid & Electromagnet

A coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder is called a **solenoid**.

- The magnetic field inside a solenoid is **uniform** (parallel straight lines).
- One end behaves like a magnetic North pole and the other as a South pole (similar to a bar magnet).
- **Electromagnet:** A strong magnetic field produced inside a solenoid can be used to magnetize a piece of soft iron core placed inside the coil. Soft iron loses its magnetism when current is turned off (temporary magnet).

3. Force on a Current-Carrying Conductor in a Magnetic Field

A current-carrying conductor experiences a mechanical force when placed in a magnetic field. The force is maximum when the conductor is placed **perpendicular (90°)** to the magnetic field and zero when parallel (0°).

Fleming's Left-Hand Rule (For Force / Motion)

Stretch the thumb, forefinger, and middle finger of your left hand mutually perpendicular to each other:

- **Forefinger:** Points in direction of *Magnetic Field*.
- **Middle Finger:** Points in direction of *Current*.
- **Thumb:** Points in direction of *Motion / Force* acting on the conductor.

4. Domestic Electric Circuits

4.1 Wire Supply System

- **Live Wire (Phase):** Red/Brown insulation, at $220 ext{ V}$ potential.
- **Neutral Wire:** Black/Blue insulation, at $0 ext{ V}$ potential. (Potential difference = $220 ext{ V}$).
- **Earth Wire:** Green/Yellow insulation, connected to a metal plate buried deep in the ground near the house.

4.2 Safety Features

- **Earthing:** Provides a low-resistance conducting path for leakage electric currents to ground, protecting users from severe electric shocks when touching metallic bodies of appliances.
- **Electric Fuse:** Safety device containing a thin wire of low melting point connected in series with live wire. Breaks the circuit if current exceeds safe limits.
- **Short-Circuiting:** Occurs when live and neutral wires come in direct contact (e.g., damaged insulation), causing circuit resistance to drop drastically and current to surge dangerously.
- **Overloading:** Occurs when too many high-power appliances are connected to a single socket simultaneously, drawing total current beyond the wire capacity.

5. Solved Examples & Practice Questions

Solved Example 5.1

Question: An electron enters a uniform magnetic field at right angles to it. The direction of magnetic field is towards the right and the electron moves vertically downwards. What is the direction of force acting on the electron?

SOLUTION:
Direction of current is taken **opposite** to the direction of motion of negative charges (electrons). Thus, electric current is directed vertically upwards.

Applying **Fleming's Left-Hand Rule**:

- Forefinger = Field (pointing Right)
- Middle finger = Current (pointing Upwards)
- Thumb points **into the page**.

Therefore, the force acts directed **into the page**.

Question 1 (Conceptual - 2 Marks)

Why do two magnetic field lines never intersect each other?

SOLUTION:
If two field lines intersected at a point, it would mean that a compass needle placed at that point would point in two different directions at the exact same time. Since a magnetic field can have only one net direction at any given point, intersection is impossible.

Question 2 (Comparison - 3 Marks)

Distinguish between a Permanent Magnet and an Electromagnet.

SOLUTION:

Feature	Electromagnet	Permanent Magnet
Magnetism	Temporary (loses magnetism when current stops)	Permanent (retains magnetism over time)
Core Material	Soft iron core	Steel or alloys like Alnico/Nipermag
Strength Control	Can be easily altered by changing current or turns	Fixed magnetic strength
Polarity	Can be reversed by reversing current direction	Fixed poles (North and South)

Question 3 (Application - 3 Marks)

What is the function of an earth wire? Why is it necessary to earth metallic appliances?

SOLUTION:
The earth wire provides a low-resistance path for electric current. When live wire insulation wears off and touches the metallic body of an appliance (e.g., electric iron, refrigerator), the appliance body gets energized.

The earth wire instantly safely diverts this fault current straight to the earth plate, preventing electric shock to any user who touches the appliance while blowing the circuit fuse due to current surge.

Question 4 (Rule Application - 2 Marks)

A current flows horizontally from East to West in a straight power line. What is the direction of the magnetic field at a point directly below it?

SOLUTION:

Applying the **Right-Hand Thumb Rule**:

Point the right thumb towards West (direction of current). Below the wire, the curled fingers point towards the **North** direction. Hence, the magnetic field directly below the wire is directed towards the **North**.